

Proof that the $O(n)$ Approach for the Diamond Heuristic in Vacuum World is Correct

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Definitions

- Let G be the set $\mathbb{Z} \times \mathbb{Z}$, with each ordered pair representing the coordinates of a square on an infinite grid.

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	(-2, 0)	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)	(-1, -1)	(0, -1)	(1, -1)	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: A 9×9 subsection of the infinite grid G .

- Let $FD\#(m)$, where m is a square $(x, y) \in G$, equal $y + x$.

0	1	2	3	4	5	6	7	8
-1	0	1	2	3	4	5	6	7
-2	-1	0	1	2	3	4	5	6
-3	-2	-1	0	1	2	3	4	5
-4	-3	-2	-1	0	1	2	3	4
-5	-4	-3	-2	-1	0	1	2	3
-6	-5	-4	-3	-2	-1	0	1	2
-7	-6	-5	-4	-3	-2	-1	0	1
-8	-7	-6	-5	-4	-3	-2	-1	0

Pictured: The $FD\#$ of each square in G .

- Let $FD(p)$, where $p \in \mathbb{Z}$, be the subset $\{m \in G : FD\#(m) = p\}$, representing an infinitely spanning falling diagonal.
- Let $FD_{square}(m)$, where m is a square $(x, y) \in G$, be $FD(FD\#(m))$, representing the falling diagonal which intersects $m = (x, y)$.

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	(-2, 0)	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)	(-1, -1)	(0, -1)	(1, -1)	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: The falling diagonal $FD_{square}(m)$, where $m = (2, 1)$.

- Let $distance(FD(p), FD(q))$ equal $|q - p|$, representing the distance between two falling diagonals.
- Let $RD\#(m)$, where m is a square $(x, y) \in G$, equal $y - x$.

8	7	6	5	4	3	2	1	0
7	6	5	4	3	2	1	0	-1
6	5	4	3	2	1	0	-1	-2
5	4	3	2	1	0	-1	-2	-3
4	3	2	1	0	-1	-2	-3	-4
3	2	1	0	-1	-2	-3	-4	-5
2	1	0	-1	-2	-3	-4	-5	-6
1	0	-1	-2	-3	-4	-5	-6	-7
0	-1	-2	-3	-4	-5	-6	-7	-8

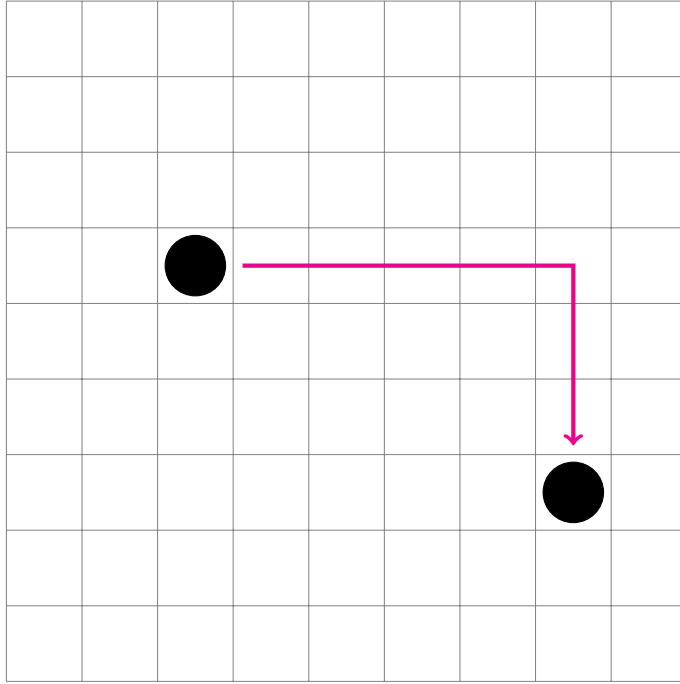
Pictured: The $RD\#$ of each square in G .

- Let $RD(p)$, where $p \in \mathbb{Z}$, be the subset $\{m \in G : RD\#(m) = p\}$, representing an infinitely spanning rising diagonal.
- Let $RD_{square}(m)$, where m is a square $(x, y) \in G$, be $RD(RD\#(m))$, representing the rising diagonal which intersects $m = (x, y)$.

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	(-2, 0)	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)	(-1, -1)	(0, -1)	(1, -1)	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: The rising diagonal $RD_{square}(m)$, where $m = (-1, 3)$.

- Let $distance(RD(p), RD(q))$ equal $|q - p|$, representing the distance between two rising diagonals.
- Let the Manhattan distance $Manhattan(m, n)$ between two squares $m = (x, y), n = (a, b)$ equal $|b - y| + |a - x|$.



Pictured: A representation of the manhattan distance between two squares.
In this case, the manhattan distance is 8.

Manhattan Diagonal Equality Lemma

Suppose we have two squares $m = (x, y), n = (a, b) \in G$.

Case 1: $a \geq x$ and $b \geq y$

Theorem: If this is the case, then it must follow that

$$\text{Manhattan}(m, n) = \text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$$

0	1	2	3	4	5	6	7	8
-1	0	1	2	3	n	5	6	7
-2	-1	0	1	2	3	4	5	6
-3	-2	-1	0	1	2	3	4	5
-4	-3	-2	-1	0	1	2	3	4
-5	-4	-3	-2	-1	0	1	2	3
-6	-5	-4	-3	-2	-1	0	1	2
-7	-6	m	-4	-3	-2	-1	0	1
-8	-7	-6	-5	-4	-3	-2	-1	0

Pictured: Example positions of $m = (x, y), n = (a, b) \in G$, where $a \geq x$ and $b \geq y$. The FD# of each square is shown. The above theorem states that the distance between the two blue falling diagonals must be the same as the Manhattan distance (magenta arrow) between m and n .

Proof:

By definition,

$$distance(FD_{square}(m), FD_{square}(n)) = distance(FD(y+x), FD(b+a))$$

Also by definition,

$$distance(FD(y+x), FD(b+a)) = |(b+a) - (y+x)|$$

Rearranging the terms,

$$|(b+a) - (y+x)| = |b+a-y-x| = |(b-y) + (a-x)|$$

We want to show that

$$Manhattan(m, n) = |(b-y) + (a-x)|$$

$$|b-y| + |a-x| = |(b-y) + (a-x)|$$

We can manipulate the inequalities given above.

$$a \geq x$$

$$a - x \geq 0$$

$$b \geq y$$

$$b - y \geq 0$$

Since we know that $a - x$ and $b - y$ must both be non-negative, we can remove the absolute value operators on the left side of the equation.

$$(b - y) + (a - x) = |(b - y) + (a - x)|$$

Adding both inequalities together, we get

$$(b - y) + (a - x) \geq 0$$

Since we know that $(b - y) + (a - x)$ must be non-negative, we can remove the absolute value operator on the right side of the equation.

$$(b - y) + (a - x) = (b - y) + (a - x)$$

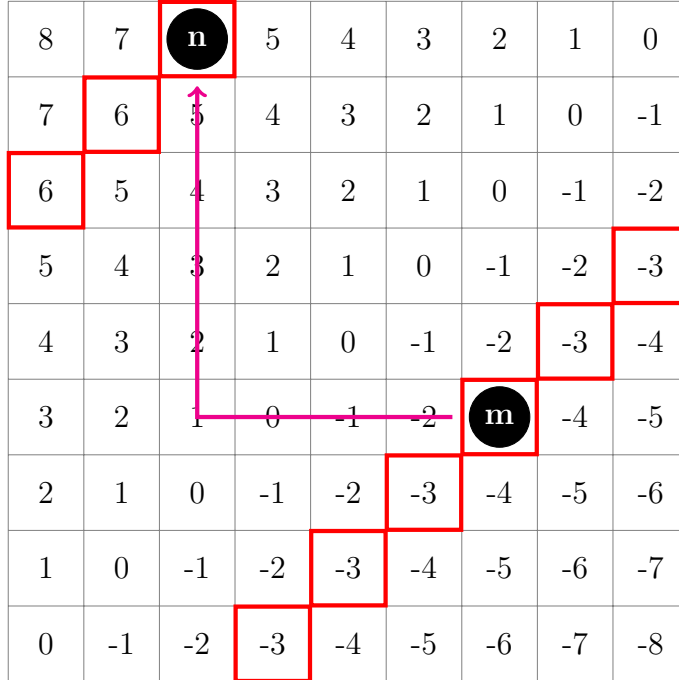
Thus, if $a > x$ and $b > y$, then

$$Manhattan(m, n) = distance(FD_{square}(m), FD_{square}(n))$$

Case 2: $x \geq a$ and $b \geq y$

Theorem: If this is the case, then it must follow that

$$Manhattan(m, n) = distance(RD_{square}(m), RD_{square}(n))$$



Pictured: Example positions of $m = (x, y), n = (a, b) \in G$, where $x \geq a$ and

$b \geq y$. The RD# of each square is shown. The above theorem states that the distance between the two red rising diagonals must be the same as the Manhattan distance (magenta arrow) between m and n .

Proof:

By definition,

$$distance(RD_{square}(m), RD_{square}(n)) = distance(RD(y-x), RD(b-a))$$

Also by definition,

$$distance(RD(y-x), RD(b-a)) = |(b-a) - (y-x)|$$

Rearranging the terms,

$$|(b-a) - (y-x)| = |b-a-y+x| = |(b-y) + (x-a)|$$

We want to show that

$$Manhattan(m, n) = |(b-y) + (x-a)|$$

$$|b-y| + |a-x| = |(b-y) + (x-a)|$$

The terms on the left side can then be shifted around to match more closely with the terms on the right.

$$|b-y| + |- (x-a)| = |(b-y) + (x-a)|$$

$$|b-y| + |x-a| = |(b-y) + (x-a)|$$

We can manipulate the inequalities given above.

$$x \geq a$$

$$x - a \geq 0$$

$$b \geq y$$

$$b - y \geq 0$$

Since we know that $x-a$ and $b-y$ must both be non-negative, we can remove the absolute value operators on the left side of the equation.

$$(b-y) + (x-a) = |(b-y) + (x-a)|$$

Adding both inequalities together, we get

$$(b-y) + (x-a) \geq 0$$

Since we know that $(b-y) + (x-a)$ must be non-negative, we can remove the absolute value operator on the right side of the equation.

$$(b-y) + (x-a) = (b-y) + (x-a)$$

Thus, if $x \geq a$ and $b \geq y$, then

$$Manhattan(m, n) = distance(RD_{square}(m), RD_{square}(n))$$

Preemptive Theorem

Suppose we have two numbers $c \in \mathbb{R}$, $d \in \mathbb{R}$.

Theorem: If $c \geq 0$ and $d \geq 0$, then

$$c + d \geq |c - d|$$

Proof:

Two possible cases must be accounted for.

Case 1: $c \geq d$

This is the same as

$$c - d \geq 0$$

Since we know that $c - d$ is non-negative, we can remove the absolute value operator on the right side of our inequality.

$$c + d \geq c - d$$

$$d \geq -d$$

$$2d \geq 0$$

$$d \geq 0$$

Thus, if $c \geq d$, then it must be true that

$$c + d \geq |c - d|$$

Case 2: $d \geq c$

This is the same as

$$d - c \geq 0$$

Our inequality can be manipulated as such:

$$c + d \geq | - (-c + d) |$$

$$c + d \geq |d - c|$$

Since we know that $d - c$ is non-negative, we can remove the absolute value operator on the right side of our inequality.

$$c + d \geq d - c$$

$$c \geq -c$$

$$2c \geq 0$$

$$c \geq 0$$

Thus, if $d \geq c$, then it must be true that

$$c + d \geq |c - d|$$

With both cases covered, we know it if $c \geq 0$ and $d \geq 0$, then it is always true that

$$c + d \geq |c - d|$$

Greater Diagonal Distance Lemma

Suppose we have two squares $m = (x, y), n = (a, b) \in G$.

Case 1: $a \geq x$ and $b \geq y$

Theorem: If this is the case, then it must follow that

$$distance(FD_{square}(m), FD_{square}(n)) \geq distance(RD_{square}(m), RD_{square}(n))$$

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	n	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	m	(-2, 0)	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)	(-1, -1)	(0, -1)	(1, -1)	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: Example positions of $m = (x, y), n = (a, b) \in G$, where $a \geq x$ and $b \geq y$. The coordinates of each square in G are shown. The above theorem states that the distance between the two blue falling diagonals must be greater than or equal to the distance between the two red rising diagonals.

Proof:

Following the simplification process shown in the previous section, we can reduce this inequality down to

$$|(b - y) + (a - x)| \geq |(b - y) + (x - a)|$$

Additionally, from the given inequalities $a \geq x$ and $b \geq y$, we get that

$$a - x \geq 0$$

$$b - y \geq 0$$

$$(b - y) + (a - x) \geq 0$$

Since we know that $(b - y) + (a - x)$ is non-negative, we can remove the absolute value operator on the left side of our original inequality.

$$(b - y) + (a - x) \geq |(b - y) + (x - a)|$$

We can then manipulate the right side of our inequality to match more closely with the left side.

$$(b - y) + (a - x) \geq |(b - y) - (a - x)|$$

Since both $b - y$ and $a - x$ are non-negative, the Preemptive Theorem says that this inequality always holds true.

Thus, if $a \geq x$ and $b \geq y$, then

$$distance(FD_{square}(m), FD_{square}(n)) \geq distance(RD_{square}(m), RD_{square}(n))$$

Case 2: $x \geq a$ and $b \geq y$

Theorem: If this is the case, then it must follow that

$$distance(RD_{square}(m), RD_{square}(n)) \geq distance(FD_{square}(m), FD_{square}(n))$$

(-4, 4)	(-3, 4)	(-2, 4)	n	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	(-2, 0)	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)	(-1, -1)	(0, -1)	(1, -1)	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	m	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: Example positions of $m = (x, y), n = (a, b) \in G$, where $x \geq a$ and $b \geq y$. The coordinates of each square in G are shown. The above theorem states that the distance between the two red rising diagonals must be greater than or equal to the distance between the two blue falling diagonals.

Proof:

Following the simplification process shown in the previous section, we can reduce this inequality down to

$$|(b - y) + (x - a)| \geq |(b - y) + (a - x)|$$

Additionally, from the given inequalities $x \geq a$ and $b \geq y$, we get that

$$x - a \geq 0$$

$$b - y \geq 0$$

$$(b - y) + (x - a) \geq 0$$

Since we know that $(b - y) + (x - a)$ is non-negative, we can remove the absolute value operator on the left side of our original inequality.

$$(b - y) + (x - a) \geq |(b - y) + (a - x)|$$

We can then manipulate the right side of our inequality to match more closely with the left side.

$$(b - y) + (x - a) \geq |(b - y) - (x - a)|$$

Since both $b - y$ and $x - a$ are non-negative, the Preemptive Theorem says that this inequality always holds true.

Thus, if $x \geq a$ and $b \geq y$, then

$$distance(RD_{square}(m), RD_{square}(n)) \geq distance(FD_{square}(m), FD_{square}(n))$$

Case 2: $RD\#(m) > RD\#(n)$

Theorem: If $RD\#(m) > RD\#(n)$ and both m and n are reflected across the origin, such that $m' = (-x, -y)$ and $n' = (-a, -b)$, then

$$RD\#(n') > RD\#(m')$$

Proof:

Suppose $RD\#(m) > RD\#(n)$. This is the same as

$$y - x > b - a$$

Multiplying both sides of this inequality by -1, we get

$$-y + x < -b + a$$

Which is equal to

$$(-y) - (-x) < (-b) - (-a)$$

Which is the same as

$$RD\#(m') < RD\#(n')$$

Thus,

$$RD\#(n') > RD\#(m')$$

Four Number Position Lemma

Suppose we have four numbers $a, b, c, d \in \mathbb{R}$, with $a - b > c - d$.

Theorem: The statements $a \leq c$ and $b \geq d$ cannot be true at the same time.

Proof:

We will use a proof by contradiction.

Suppose $a \leq c$ and $b \geq d$. This gives us

$$a - c \leq 0$$

$$b - d \geq 0$$

We will now rearrange the inequality $a - b > c - d$ into

$$a - c > b - d$$

Since $a - c$ is non-positive and $b - d$ is non-negative, this is impossible. Due to this contradiction, $a \leq c$ and $b \geq d$ cannot be true simultaneously.

Furthest Diagonals Lemma

Suppose we have a subset $S \subset G$ which contains two or more elements. Let $m = (x, y), n = (a, b)$ be squares such that $m, n \in S$.

Case 1: $\min_{u \in S}(FD\#(u)) = FD\#(m)$ and $\max_{u \in S}(FD\#(u)) = FD\#(n)$

Theorem: If this is the case, then it must follow that

$$\max_{u, v \in S}(\text{distance}(FD_{\text{square}}(u), FD_{\text{square}}(v))) = \text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$$

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	●	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	●	(1, 1)	(2, 1)	n	(4, 1)
(-4, 0)	(-3, 0)	●	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)	(-1, -1)	(0, -1)	(1, -1)	●	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	●	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	m	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	●	(4, -4)

Pictured: All of the squares in an example S , including examples of m and n . The coordinates of each square in G are shown. The falling diagonals of m and n are shown in blue, with m having the minimum $FD\#$ in S and n having the maximum $FD\#$ in S . The above theorem states that the distance between these two falling diagonals is the greatest falling diagonal distance out of any

two squares in S .

Proof:

We will use a proof by contradiction.

Suppose $\min_{u \in S}(FD\#(u)) = FD\#(m)$ and $\max_{u \in S}(FD\#(u)) = FD\#(n)$. Also suppose that

$$\max_{u,v \in S}(\text{distance}(FD_{\text{square}}(u), FD_{\text{square}}(v))) \neq \text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$$

This means that there must exist squares $i, j \in S$ such that

$$\text{distance}(FD_{\text{square}}(i), FD_{\text{square}}(j)) > \text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$$

This is the same as

$$|FD\#(j) - FD\#(i)| > |FD\#(n) - FD\#(m)|$$

We will assume that $FD\#(n) \geq FD\#(m)$ and $FD\#(j) \geq FD\#(i)$. This yields

$$FD\#(n) - FD\#(m) \geq 0$$

$$FD\#(j) - FD\#(i) \geq 0$$

Since both $FD\#(j) - FD\#(i)$ and $FD\#(n) - FD\#(m)$ are non-negative, we can get rid of the absolute value symbols in our original inequality.

$$FD\#(j) - FD\#(i) > FD\#(n) - FD\#(m)$$

Given this inequality, the Four Number Position Lemma says that it's impossible for both $FD\#(j) \leq FD\#(n)$ and $FD\#(i) \geq FD\#(m)$ to be true simultaneously. Therefore, either $FD\#(j) > FD\#(n)$ or $FD\#(i) < FD\#(m)$ must be true. However, this contradicts the definition of m and n : they no longer have the min and max $FD\#$ respectively. Since the statement

$$\max_{u,v \in S}(\text{distance}(FD_{\text{square}}(u), FD_{\text{square}}(v))) \neq \text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$$

gives us a contradiction, it must instead be true that

$$\max_{u,v \in S}(\text{distance}(FD_{\text{square}}(u), FD_{\text{square}}(v))) = \text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$$

Case 2: $\min_{u \in S}(RD\#(u)) = RD\#(m)$ and $\max_{u \in S}(RD\#(u)) = RD\#(n)$

Theorem: If this is the case, then it must follow that

$$\max_{u,v \in S}(\text{distance}(RD_{\text{square}}(u), RD_{\text{square}}(v))) = \text{distance}(RD_{\text{square}}(m), RD_{\text{square}}(n))$$

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	n	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	●	(-1, 1)	●	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	(-2, 0)	(-1, 0)	(0, 0)	(1, 0)	(2, 0)	(3, 0)	●
(-4, -1)	(-3, -1)	(-2, -1)	●	(0, -1)	(1, -1)	(2, -1)	(3, -1)	●
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	●	(2, -2)	(3, -2)	(4, -2)
●	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	m	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: All of the squares in an example S , including examples of m and n . The coordinates of each square in G are shown. The rising diagonals of m and n are shown in red, with m having the minimum RD# in S and n having the maximum RD# in S . The above theorem states that the distance between these two rising diagonals is the greatest rising diagonal distance out of any two squares in S .

Proof:

We will use a proof by contradiction.

Suppose $\min_{u \in S}(RD\#(u)) = RD\#(m)$ and $\max_{u \in S}(RD\#(u)) = RD\#(n)$. Also suppose that

$$\max_{u, v \in S}(\text{distance}(RD_{\text{square}}(u), RD_{\text{square}}(v))) \neq \text{distance}(RD_{\text{square}}(m), RD_{\text{square}}(n))$$

This means that there must exist squares $i, j \in S$ such that

$$\text{distance}(RD_{\text{square}}(i), RD_{\text{square}}(j)) > \text{distance}(RD_{\text{square}}(m), RD_{\text{square}}(n))$$

This is the same as

$$|RD\#(j) - RD\#(i)| > |RD\#(n) - RD\#(m)|$$

We will assume that $RD\#(n) \geq RD\#(m)$ and $RD\#(j) \geq RD\#(i)$. This yields

$$RD\#(n) - RD\#(m) \geq 0$$

$$RD\#(j) - RD\#(i) \geq 0$$

Since both $RD\#(j) - RD\#(i)$ and $RD\#(n) - RD\#(m)$ are non-negative, we can get rid of the absolute value symbols in our original inequality.

$$RD\#(j) - RD\#(i) > RD\#(n) - RD\#(m)$$

Given this inequality, the Four Number Position Lemma says that it's impossible for both $RD\#(j) \leq RD\#(n)$ and $RD\#(i) \geq RD\#(m)$ to be true simultaneously. Therefore, either $RD\#(j) > RD\#(n)$ or $RD\#(i) < RD\#(m)$ must be true. However, this contradicts the definition of m and n : they no longer have the min and max $RD\#$ respectively. Since the statement

$$\max_{u,v \in S} (\text{distance}(RD_{\text{square}}(u), RD_{\text{square}}(v))) \neq \text{distance}(RD_{\text{square}}(m), RD_{\text{square}}(n))$$

gives us a contradiction, it must instead be true that

$$\max_{u,v \in S} (\text{distance}(RD_{\text{square}}(u), RD_{\text{square}}(v))) = \text{distance}(RD_{\text{square}}(m), RD_{\text{square}}(n))$$

Manhattan Subset Lemma

Theorem: For any subset $S \subset G$ with two or more squares, it must be true that $\{\text{Manhattan}(u, v) | u, v \in S\} \subset \{\text{distance}(FD_{\text{square}}(u), FD_{\text{square}}(v)) | u, v \in S\} \cup \{\text{distance}(RD_{\text{square}}(u), RD_{\text{square}}(v)) | u, v \in S\}$.

Proof:

Suppose we have a subset $S \subset G$ with at least two squares. Additionally, suppose we have two squares $m = (x, y), n = (a, b) \in S$. We will look at each of their possible orientations.

If $b \geq y$ and $a \geq x$, then Case 1 of the Manhattan Diagonal Equality Lemma says that

$$\text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n)) = \text{Manhattan}(m, n)$$

If $b \geq y$ and $x \geq a$, then Case 2 of the Manhattan Diagonal Equality Lemma says that

$$\text{distance}(RD_{\text{square}}(m), RD_{\text{square}}(n)) = \text{Manhattan}(m, n)$$

If $y \geq b$ and $a \geq x$, then Case 2 of the Manhattan Diagonal Equality Lemma says that

$$\text{distance}(RD_{\text{square}}(n), RD_{\text{square}}(m)) = \text{Manhattan}(n, m)$$

And if $y \geq b$ and $x \geq a$, then Case 1 of the Manhattan Diagonal Equality Lemma says that

$$\text{distance}(FD_{\text{square}}(n), FD_{\text{square}}(m)) = \text{Manhattan}(n, m)$$

Therefore, for any two squares $m, n \in S$, either $distance(FD_{square}(m), FD_{square}(n)) = Manhattan(m, n)$ or $distance(RD_{square}(m), RD_{square}(n)) = Manhattan(m, n)$ must be true. Thus, $\{Manhattan(u, v) | u, v \in S\} \subset \{distance(FD_{square}(u), FD_{square}(v)) | u, v \in S\} \cup \{distance(RD_{square}(u), RD_{square}(v)) | u, v \in S\}$.

Diamond Distance Theorem

Suppose we have a subset $S \subset G$ with at least two squares. Additionally, suppose we have four squares $m = (x, y), n = (a, b), o = (r, s), p = (c, d) \in S$ with $\min_{u \in S}(FD\#(u)) = FD\#(m), \max_{u \in S}(FD\#(u)) = FD\#(n), \min_{u \in S}(RD\#(u)) = RD\#(o)$, and $\max_{u \in S}(RD\#(u)) = RD\#(p)$. We want to show that $\max_{u, v \in S}(Manhattan(u, v)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

We will prove this across four different cases.

Case 1: $distance(FD_{square}(m), FD_{square}(n)) \neq Manhattan(m, n)$
and $distance(RD_{square}(o), RD_{square}(p)) \neq Manhattan(o, p)$

Theorem: It is impossible for both of these statements to be true simultaneously.

Proof:

We will use a proof by contradiction.

Case 1.1: First, let's look at the case where

$$distance(FD_{square}(m), FD_{square}(n)) \geq distance(RD_{square}(o), RD_{square}(p))$$

The Manhattan Subset Lemma shows that either $distance(FD_{square}(m), FD_{square}(n)) = Manhattan(m, n)$ or $distance(RD_{square}(m), RD_{square}(n)) = Manhattan(m, n)$ must be true. Since the former is false, the latter must therefore be true. Because the latter is true, the Manhattan Subset Lemma shows that either $b \geq y$ and $x \geq a$ or $y \geq b$ and $a \geq x$ must be true. In either case, Case 2 of the Greater Diagonal Distance Lemma says that

$$distance(RD_{square}(m), RD_{square}(n)) \geq distance(FD_{square}(m), FD_{square}(n))$$

Combined with the previous inequality, this gives us

$$distance(RD_{square}(m), RD_{square}(n)) \geq distance(RD_{square}(o), RD_{square}(p))$$

Using Case 2 of the Furthest Diagonals Lemma, we can change the right side of this inequality.

$$distance(RD_{square}(m), RD_{square}(n)) \geq \max_{u, v \in S}(distance(RD_{square}(u), RD_{square}(v)))$$

This is a contradiction, which will soon be important.

Case 1.2: Next, let's look at the case where

$$distance(RD_{square}(o), RD_{square}(p)) \geq distance(FD_{square}(m), FD_{square}(n))$$

The Manhattan Subset Lemma shows that either $distance(FD_{square}(o), FD_{square}(p)) = Manhattan(o, p)$ or $distance(RD_{square}(o), RD_{square}(p)) = Manhattan(o, p)$ must be true. Since the latter is false, the former must therefore be true. Because the former is true, the Manhattan Subset Lemma shows that either $d \geq s$ and $c \geq r$ or $s \geq d$ and $r \geq c$ must be true. In either case, Case 1 of the Greater Diagonal Distance Lemma says that

$$distance(FD_{square}(o), FD_{square}(p)) \geq distance(RD_{square}(o), RD_{square}(p))$$

Combined with the previous inequality, this gives us

$$distance(FD_{square}(o), FD_{square}(p)) \geq distance(FD_{square}(m), FD_{square}(n))$$

Using Case 1 of the Furthest Diagonals Lemma, we can change the right side of this inequality.

$$distance(FD_{square}(o), FD_{square}(p)) \geq \max_{u,v \in S}(distance(FD_{square}(u), FD_{square}(v)))$$

This is another contradiction.

Because of these two contradictions, neither the statement

$$distance(FD_{square}(m), FD_{square}(n)) \geq distance(RD_{square}(o), RD_{square}(p))$$

nor the statement

$$distance(RD_{square}(o), RD_{square}(p)) \geq distance(FD_{square}(m), FD_{square}(n))$$

can be true. It should be impossible for neither of them to be true, which proves that there is never a situation where both $distance(FD_{square}(m), FD_{square}(n)) \neq Manhattan(m, n)$ and $distance(RD_{square}(o), RD_{square}(p)) \neq Manhattan(o, p)$.

Case 2: $distance(FD_{square}(m), FD_{square}(n)) \neq Manhattan(m, n)$
and $distance(RD_{square}(o), RD_{square}(p)) = Manhattan(o, p)$

Theorem: Given these conditions, it must be true that $\max_{u,v \in S}(Manhattan(u, v)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	p	(-1, 2)	(0, 2)	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	n	(-2, 0)	(-1, 0)	(0, 0)		(2, 0)	(3, 0)	(4, 0)
(-4, -1)	(-3, -1)	(-2, -1)		(0, -1)	(1, -1)	m	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)		(0, -2)		(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	(-1, -3)	(0, -3)	(1, -3)	o	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: All of the squares in an example S , including examples of m , n , o , and p . The coordinates of each square in G are shown. The falling diagonals of m and n are shown in blue, with m having the minimum FD# in S and n having the maximum FD# in S . The rising diagonals of o and p are shown in red, with o having the minimum RD# in S and p having the maximum RD# in S . m and n have the highest falling diagonal distance out of all of the squares in S , despite that distance not being their actual manhattan distance. The above theorem states that if the FD distance between m and n and the RD distance between o and p are found, and the maximum between the two values is taken, then the resulting number is the same as the highest manhattan distance between any two squares in S . The diagonal lines intersect each other to create a diamond-like shape, hence the name of the theorem.

Proof:

First, we must show that

$$distance(RD_{square}(o), RD_{square}(p)) > distance(FD_{square}(m), FD_{square}(n))$$

For this, we will use a proof by contradiction. Suppose that

$$distance(FD_{square}(m), FD_{square}(n)) \geq distance(RD_{square}(o), RD_{square}(p))$$

Case 1.1 of the Diamond Distance Theorem shows that this statement, combined with the given conditions, leads to a contradiction. Thus,

$$distance(RD_{square}(o), RD_{square}(p)) > distance(FD_{square}(m), FD_{square}(n))$$

Because of this, we now have $distance(RD_{square}(o), RD_{square}(p)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$. This is the same as $Manhattan(o, p) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$. We can also change the right side of the inequality using Cases 1 and 2 of the Furthest Diagonals Lemma: $Manhattan(o, p) = \max(\max_{u,v \in S}(distance(FD_{square}(u), FD_{square}(v))), \max_{u,v \in S}(distance(RD_{square}(u), RD_{square}(v))))$. Because the Manhattan Subset Lemma says that the set of all manhattan distances in S is a subset of the set of all FD and RD distances in S , and the previous equation shows that the max operation returns a value which is present within that set of manhattan distances, it must therefore be true that $\max_{u,v \in S}(Manhattan(u, v)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

Case 3: $distance(FD_{square}(m), FD_{square}(n)) = Manhattan(m, n)$
and $distance(RD_{square}(o), RD_{square}(p)) \neq Manhattan(o, p)$

Theorem: Given these conditions, it must be true that $\max_{u,v \in S}(Manhattan(u, v)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	(0, 2)	●	n	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	(-2, 1)	(-1, 1)	(0, 1)	(1, 1)	(2, 1)	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	●	(-1, 0)	(0, 0)	(1, 0)	●	p	(4, 0)
(-4, -1)	o	(-2, -1)	(-1, -1)	●	(1, -1)	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	(-1, -2)	(0, -2)	(1, -2)	(2, -2)	(3, -2)	(4, -2)
(-4, -3)	(-3, -3)	m	(-1, -3)	(0, -3)	(1, -3)	(2, -3)	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: The same type of content as the last picture, except that o and p have the highest rising diagonal distance out of all of the squares in S , despite that distance not being their actual manhattan distance.

Proof:

First, we must show that

$$distance(FD_{square}(m), FD_{square}(n)) > distance(RD_{square}(o), RD_{square}(p))$$

For this, we will use a proof by contradiction. Suppose that

$$distance(RD_{square}(o), RD_{square}(p)) \geq distance(FD_{square}(m), FD_{square}(n))$$

Case 1.2 of the Diamond Distance Theorem shows that this statement, combined with the given conditions, leads to a contradiction. Thus,

$$distance(FD_{square}(m), FD_{square}(n)) > distance(RD_{square}(o), RD_{square}(p))$$

Because of this, we now have $distance(FD_{square}(m), FD_{square}(n)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$. This is the same as $Manhattan(m, n) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

We can also change the right side of the inequality using Cases 1 and 2 of the Furthest Diagonals Lemma: $Manhattan(m, n) = \max(\max_{u,v \in S}(distance(FD_{square}(u), FD_{square}(v))), \max_{u,v \in S}(distance(RD_{square}(u), RD_{square}(v))))$. Because the Manhattan Subset Lemma says that the set of all manhattan distances in S is a subset of the set of all FD and RD distances in S , and the previous equation shows that the max operation returns a value which is present within that set of manhattan distances, it must therefore be true that $\max_{u,v \in S}(Manhattan(u, v)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

Case 4: $distance(FD_{square}(m), FD_{square}(n)) = Manhattan(m, n)$
and $distance(RD_{square}(o), RD_{square}(p)) = Manhattan(o, p)$

Theorem: Given these conditions, it must be true that $\max_{u,v \in S}(Manhattan(u, v)) = \max(distance(FD_{square}(m), FD_{square}(n)), distance(RD_{square}(o), RD_{square}(p)))$.

(-4, 4)	(-3, 4)	(-2, 4)	(-1, 4)	(0, 4)	(1, 4)	(2, 4)	(3, 4)	(4, 4)
(-4, 3)	(-3, 3)	(-2, 3)	(-1, 3)	(0, 3)	(1, 3)	(2, 3)	(3, 3)	(4, 3)
(-4, 2)	(-3, 2)	(-2, 2)	(-1, 2)	●	(1, 2)	(2, 2)	(3, 2)	(4, 2)
(-4, 1)	(-3, 1)	p	(-1, 1)	(0, 1)	(1, 1)	n	(3, 1)	(4, 1)
(-4, 0)	(-3, 0)	(-2, 0)	(-1, 0)	●	(1, 0)	(2, 0)	(3, 0)	(4, 0)
(-4, -1)	●	(-2, -1)	(-1, -1)	(0, -1)	●	(2, -1)	(3, -1)	(4, -1)
(-4, -2)	(-3, -2)	(-2, -2)	●	(0, -2)	(1, -2)	(2, -2)	o	(4, -2)
(-4, -3)	(-3, -3)	(-2, -3)	m	(0, -3)	(1, -3)	●	(3, -3)	(4, -3)
(-4, -4)	(-3, -4)	(-2, -4)	(-1, -4)	(0, -4)	(1, -4)	(2, -4)	(3, -4)	(4, -4)

Pictured: The same type of content as the last picture, except that the falling diagonal distance between m and n , as well as the rising diagonal distance between o and p , reflect their actual manhattan distance.

Proof:

From the conditions, we already know that the result of $\max(\text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n)), \text{distance}(RD_{\text{square}}(o), RD_{\text{square}}(p)))$ will be a value that exists in the set $\{\text{Manhattan}(u, v) | u, v \in S\}$. Furthermore, we can change the expression using Cases 1 and 2 of the Furthest Diagonals Lemma, yielding $\max(\max_{u, v \in S}(\text{distance}(FD_{\text{square}}(u), FD_{\text{square}}(v))), \max_{u, v \in S}(\text{distance}(RD_{\text{square}}(u), RD_{\text{square}}(v))))$. Because of the Manhattan Subset Lemma, we know that the result of this expression won't just be a manhattan distance, but rather the greatest manhattan distance between any two squares in S . This finally gives us $\max_{u, v \in S}(\text{Manhattan}(u, v)) = \max(\text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n)), \text{distance}(RD_{\text{square}}(o), RD_{\text{square}}(p)))$.

Final Statement

These four cases show that regardless of whether $\text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n))$ or $\text{distance}(RD_{\text{square}}(o), RD_{\text{square}}(p))$ can be represented as $\text{Manhattan}(m, n)$ and $\text{Manhattan}(o, p)$ respectively, it will always hold that $\max_{u, v \in S}(\text{Manhattan}(u, v)) = \max(\text{distance}(FD_{\text{square}}(m), FD_{\text{square}}(n)), \text{distance}(RD_{\text{square}}(o), RD_{\text{square}}(p)))$.